

Some Popular Inverters with UPS

Company	Type/model	Capacity	Price (Rs)	Features
Luminous	Sinewave long backup UPS	600VA 800VA	6900 8950	Very high cold start by the UPS, fast changeover time, surface mount technology for reliable and compact design, sensor charge technology for faster charging and longer battery life, intelligent thermal management, free pre-loaded battery management software, miniature circuit breaker instead of fuse to protect load and equipments, single battery, 10-12 hours typical recharge time, low battery and reverse polarity protection, LED display for switch on, UPS on, low battery pre-alarm, battery low, mains on, smart charge, overload, short circuit under battery mode, and MCB trip or short circuit under mains mode, alarms for low battery pre-alarm, low battery, overload, no load shut down and short circuit, 0-40°C operating temperature
Microtek	Model UPS E ²	625VA	4290	Highly energy efficient, in-built low watt loss transformer saves the power while charging the battery, two years warranty, single-battery system, TPZI waveform, 110V-310V input AC range, 10.5-14.2 input DC range, overload protection (110 per cent), short-circuit protection (300 per cent), charger current protection (6.75A)
	Model UPS EB 850	850VA	4990	Single-battery system, TPZI waveform, 110V-310V input AC range, 10.5-14.2 input DC range, overload protection (110 per cent), short-circuit protection (300 per cent), charger current protection (9.00A)
Parker Power	Pure sinewave solar inverter/ model SUN	810VA/ 12V DC	—	Dual charging options (solar as well as mains utility), 100 per cent pure sinewave output, dual functionality modes (inverter and UPS), with UPS mode, computers can be directly connected to UPS without the need of a separate UPS for each computer, smart charge pulse technology and solar charge controller, advanced thermal management for longer life and higher reliability of inverter, high-end microcontroller/digital signal controller based design, unique overload and short circuit protection with auto reset feature, unique child lock protection, user-friendly graphic display, compatible with all types of batteries, four modes provided for different top-charging level, one battery
upsINVERTER.com	Sinewave Combo UPS/ model UTL-Hi-Lite	810VA	5990	Starts from as low as 90V, just 2.5 kg of total weight, 20kHz PWM based pure sine wave output, selectable fast/normal charging, auto-restartable no-load shutdown, higher efficiency that gives 15 per cent more back-up, auto-adjustment with any kind of battery, touch switch for smooth operation

Note: While every precaution has been taken to ensure accuracy and adequate information in this table, EFY assumes no responsibility and disclaims all liability for damages resulting from use of this information or for any errors or omissions. Also, prices may vary from dealer to dealer and change with time.

SMF batteries that are available in the market today are generally not the correct solution for inverter application because inverter is a heavy-duty application and SMF batteries won't give a good life for such an application.

Some companies have recently designed an SMF battery with the characteristics of thick-, flat-plate batteries. So it's a thick-plate battery which is specially designed for inverter applications. It gives you two advantages: It doesn't require topping up and can last long enough for an inverter kind of application.

The first lot of these batteries is available in the market and its acceptance has been good. Price is 25-30 per cent higher than of normal thick-plate

battery.

There are tubular batteries in normal-size container, jumbo container, tall container, etc. Capacities are generally 135 Ah, 150 Ah or 180 Ah. The jumbo and tall containers have extra electrolyte, so the topping-up frequency is reduced.

Which inverter-battery combination will work the best?

You may think a combination of the best inverter brand and the best battery brand will give you the best backup. But, rather than buying inverter and battery separately, go for a package. Otherwise, you will be running after two companies if something goes wrong with the battery

or inverter. The inverter maker will say it's a battery problem and the battery maker will say it's an inverter problem. There is no one-point solution. So buy both the battery and the inverter of the same brand, or at least from the same vendor.

Bhalla says that Luminous designs an inverter or charger for that keeping the batteries in mind. Similarly, it designs a battery keeping the inverter in mind. Since inverters and batteries are not designed in isolation, this gives the best solution to customers.

The life of a battery depends on the quality of charging by the inverter.

Battery costs one-and-a-half times the inverter. So the company should be able to take care of both—the inverter and the battery.

Can I use an automotive battery?

Automotive batteries can be used for inverters, but these are not recommended. They are not designed to withstand long discharge periods with inverters. They are designed with thin plates to deliver a high current for a very short duration, and are mainly used for starting, light-



Parker Power Sun 810 solar inverter